The Economics of Viable Continuation

Production, Obligation, Institutions,
and the Organization of Feasible Futures

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Preface

This book develops an economic theory in which the central object is the set of futures that an economy keeps open. Production, maintenance, obligation, exchange, and institutional design are studied by asking what continuations each arrangement makes possible, to whom, and at what cost to the conditions that later activity requires. The same transaction can produce income in the present while consuming capacities elsewhere, and the theory is built to make that difference visible and testable.

The order of presentation is deliberate. Parts I through VIII develop the economic concepts in their own terms, with motivating situations, worked examples, and identifiable consequences. Part IX consolidates the formal machinery and states the book's principal results with their hypotheses. Parts X, XI, and XII then give categorical, type theoretic, and sheaf theoretic interpretations. Each interpretation recovers structure already established in the economic theory and states what it preserves and what it leaves out. No chapter treats a mathematical analogy as a result.

Every chapter follows a common pattern: a motivating economic situation, definitions, a developed argument, a worked example, propositions with proofs where warranted, empirical questions, and exercises. Formal results are consequences of stipulated models. Claims about actual economies are labelled as such and are the subject of the empirical questions. The book's central research commitment is to determine when continuation-based descriptions yield explanations or decisions that ordinary accounting, optimization, and institutional analysis would miss, and to say plainly when they do not.

Draft note. Scope. The outline does not yet treat money and banking, labour markets, international trade, long-run growth, or mechanism design as separate topics. Either state this scope explicitly here, or fold labour into the chapters on exit and care, and mechanism design into the chapters on exchange and protocols. A possible additional chapter would recover Arrow–Debreu equilibrium, Solow growth, and dynamic programming as limiting cases (history compresses, viability constraints are slack, valuation is scalar), giving graduate readers a benchmark for later claims.

Draft note. Length. Sixty chapters is on the order of a thousand pages. Reading paths are given in the next section. Parts X to XII could be issued as a second volume if the length becomes unwieldy.

Reading Paths and Dependencies

The twelve parts are not meant to be read in a single sitting. Three routes through the book are offered. The economics route covers Parts I through VIII and needs no category theory, type theory, or sheaf theory. The formal-core route takes Parts I to III for the economic foundation, then Part IX for the principal results, and then one of the interpretation parts, X, XI, or XII. The full route reads the parts in order. Table 1 marks the parts each route includes, and Figure 1 shows which parts each part presupposes.

Route	I	II	III	IV	V	VI	VII	VIII	IX	X	XI	XII
Economics	■	■	■	■	■	■	■	■				
Formal core	■	■	■						■	one of X, XI, XII		
Full	■	■	■	■	■	■	■	■	■	■	■	■

Table 1: Reading routes. Shaded cells are read.

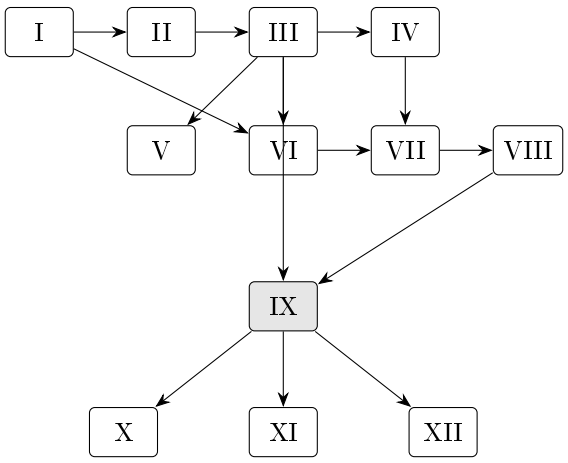


Figure 1: Dependencies among parts. An arrow from A to B means B presupposes A. Part IX is formally self-contained given Parts I to III, and the arrow from Part VIII records that its results are motivated by the whole economic text.

Housing as the running case

A housing-maintenance system recurs throughout. It begins with physical deterioration and tenant obligations, and then carries the argument through accounting, exit capacity, contractor composition, typed repair commitments, and the compatibility of local records. Each part opens with a short paragraph marked *Housing thread* stating what the case contributes there. Reading these paragraphs in order shows what each additional mathematical language adds to the same example.

Common Model and Notation

A single model is used throughout, and each chapter states any departure from it. Let $s \in S$ be an augmented economic state containing material resources, relevant institutional permissions, and outstanding obligations. In a state s the available actions are $a \in A(s)$, and the allowed disturbances are $w \in W(s, a)$. Successor states are given by

$$s' = T(s, a, w).$$

Both $A(s)$ and $W(s, a)$ are assumed nonempty unless a result says otherwise. Let $G \subseteq S$ be the set of acceptable states.

A *policy* is nonanticipating: it may use the observed history but cannot depend on disturbances that have not yet been observed. Unless stated, observation is complete, so the state and the past disturbances are observed. A policy keeps the system *in G forever* against a class of disturbances if every trajectory it generates under an allowed disturbance sequence stays in G.

Three structures are kept separate throughout: *admissibility* (which trajectories satisfy the stated obligations), *welfare* (which trajectories are better for whom), and *financial value* (how trajectories are priced or accounted). A result about one is never silently a result about another.

Symbol	Meaning
$S,\ A(s),\ W(s,a),\ T$	states, actions, disturbances, transition map
G	acceptable states
Φ	robust predecessor operator, $\Phi(X) = \{s \in G : \exists a\ \forall w,\ T(s,a,w) \in X\}$
$K = K(G)$	viability kernel: greatest fixed point of Φ
$L(h)$	set of obligation-satisfying future sequences after history h
$h \sim h'$	continuation equivalence, $L(h) = L(h')$
$F : S \to Y$	an accounting or other representation
$\preceq$	partial order on continuation profiles

The Housing Case

A single small housing economy runs through the book. It is introduced here in a deterministic version that can be calculated by hand, and later chapters add disturbances, shared resources, information problems, and institutional change to the same case with the same units, timing, and signs. Amounts are in currency units, time is in months, and nothing is discounted until a chapter says otherwise.

Participants and what is observed

A *tenant* holds liquid resources $x \geq 0$ and pays rent r per month. After rent and living costs the tenant's net monthly surplus on a monthly contract is $e \geq 0$. A *landlord* owns the dwelling and may offer contracts. A *contractor* and a *public authority* enter in later chapters, as do financial accounts and outside housing opportunities. Here the tenant's x, e, and the exit cost below are observed by the tenant. The landlord observes the contract terms and whether rent is paid.

The decision

Suppose a landlord offers a prepaid lease. The tenant pays $P = (1 - \delta)\, rT$ at the start for a lease of T months at a discount $\delta \in [0, 1)$. The prepayment is non-refundable, and the tenant pays no further rent during the lease. The landlord saves a collection and arrears cost $\kappa > 0$ per prepaid contract. The tenant must hold $x \geq P$ to be eligible.

The offer looks attractive on its financial face. The discount δrT is a transfer from landlord to tenant, so aggregate surplus rises by κ, and if $\delta rT \leq \kappa$ the landlord's profit does not fall. The difficulty is that a tenant who prepays has less cash on hand for the next H months, which is the window in which a job loss, a repair failure, or a safety problem might force a move. A move at month $t \leq H$ needs resources of at least c at that date, where c covers moving costs, a new deposit, and a first month's rent.

Definition 0.1 (Executable exit)**.** An exit within horizon H is *executable* for a tenant if there is a date $t \in \{0, \ldots, H\}$ at which the tenant's resources, with every earlier balance nonnegative, are at least c. This is the constrained reachability of P08 in the present model.

On a monthly contract with $e \geq 0$, resources at month t are $x + te$, nondecreasing, so the exit is executable exactly when $x + He \geq c$. On the prepaid lease, resources at month t are $x - P + t(e + r)$ because rent is no longer paid, and the exit is executable exactly when $x - P + H(e + r) \geq c$.

Theorem H1 (P28 in the housing model)**.** *Fix r, T, δ, e, c, H with $e \geq 0$ and $T > H$. Among tenants eligible for the prepaid lease:*

(a) Aggregate surplus, counted as the sum of landlord profit and tenant flow payoff with the tenant's stay decision held fixed, rises by κ. Landlord profit changes by $\kappa - \delta rT$ and the tenant's by δrT.

(b) The tenants who had an executable exit on the monthly contract and lose it on the prepaid lease are exactly those with

$$\max\{P,\ c - He\} \ \leq \ x \ < \ c - He + P - rH.$$

(c) This set is nonempty if and only if $H < (1 - \delta)T$ and $c > H(e + r)$.

Proof. (a) The discount is paid by the landlord and received by the tenant, so it cancels in the sum, leaving the cost saving κ. (b) A tenant is eligible if $x \geq P$. The exit was executable before if $x \geq c - He$. It is not executable after if $x - P + H(e + r) < c$, that is $x < c - He + P - rH$. Intersecting the three conditions gives the interval. (c) The interval is nonempty if and only if its upper end exceeds both lower ends. Exceeding $c - He$ requires $P - rH > 0$, that is $H < (1 - \delta)T$. Exceeding P requires $c - He - rH > 0$, that is $c > H(e + r)$. □

Hypotheses. The conclusion is carried by two assumptions: the prepayment is nonrefundable, and the exit cost c must be met from liquid resources within the window. Nonnegativity of e only simplifies the safety check, since otherwise the balance path must be tracked. The assumption most likely to fail in applications is that the surplus measure ignores the value of the exit option. If it were priced, the sign of the aggregate change would depend on the probability of needing to exit.

Worked numbers. Take $r = 1000$, $T = 12$, $\delta = 0.05$, $e = 200$, $c = 4000$, $H = 3$, $\kappa = 800$. Then $P = 11{,}400$ and $\delta rT = 600$. Aggregate surplus rises by 800, the landlord gains 200, and a tenant who never needs to move gains 600. On the monthly contract the exit is executable for $x \geq 4000 - 600 = 3400$. On the prepaid lease it is executable for $x \geq 4000 - 11{,}400 + 3 \cdot 1200 = 11{,}800$ among eligible tenants. Tenants with $11{,}400 \leq x < 11{,}800$ are eligible, would have been able to exit, and can no longer do so. Both conditions of (c) hold, since $3 < 11.4$ and $4000 > 3 \cdot 1200 = 3600$.

Interpretation. The financial improvement and the loss of exit capacity are properties of different structures, as the book's separation of welfare, admissibility, and financial value requires. The result does not say prepaid leases harm tenants in general. It gives the exact condition under which a gain in surplus coexists with a smaller safe exit set, and the exact set of tenants affected.

When compensation restores the lost capability

Suppose the landlord compensates an affected tenant with a transfer m delivered at month τ. An outside unit can be taken only within a window of months $[a, E]$, so the exit date t must satisfy $a \le t \le \min\{H, E\} =: \bar{t}$. The transfer counts only at dates $t \ge \tau$.

Theorem H2 (P29 in the housing model)**.** *Let a tenant be eligible with a positive shortfall,*

$$m^*(x) \;=\; c - (x - P) - \bar{t}\,(e + r) \;>\; 0.$$

The transfer m delivered at τ restores an executable exit if and only if all three hold: (i) availability, $a \le \bar{t}$; (ii) timing, $\tau \le \bar{t}$; (iii) sufficiency, $m \ge m^*(x)$. If (i) or (ii) fails, no amount of money restores the capability.

Proof. Resources at date t are $x - P + t(e + r) + m\,\mathbf{1}[\tau \le t]$, which is nondecreasing in t when $e + r > 0$. If $\tau \le \bar{t}$ the best date is $\bar{t}$ and the condition is $m \ge m^*(x)$. If $\tau > \bar{t}$ the transfer never counts and the shortfall stays positive. If $a > \bar{t}$ there is no admissible date. □

Hypotheses and interpretation. The three conditions correspond to the three ways money can fail: the thing to be bought may not exist in the window, the money may arrive too late, or it may be too small. A transfer that increases holdings without satisfying (i) to (iii) has not restored the capability. In the worked numbers, a tenant with $x = 11{,}500$ has $m^* = 4000 - 100 - 3 \cdot 1200 = 300$. A transfer of 300 delivered by month 3 restores the exit. A transfer of 10^6 delivered at month 4 does not. The largest shortfall among affected tenants is $m^*(P) = c - \bar{t}(e + r) = 400$, which exceeds the landlord's net gain $\kappa - \delta r T = 200$, so in this example the surplus gain does not by itself finance full restoration. That comparison is a property of these numbers, not a theorem.

Scope. Both results are proved only in this deterministic model. The general statements remain specifications (P28, P29). The companion reproduces the closed forms by brute-force enumeration over integer parameter sets, as a reported check.

Next. Part I introduces the states and histories that let this case be stated without the special structure used here. The chapter on exit (Chapter 29) generalizes Theorem H1, and the chapter on restoration obligations (Chapter 19) generalizes Theorem H2.

The Proof Programme

The results below are theorem *specifications* for the writing programme. Several elementary results already admit the short proofs sketched here. The broader economic interpretations depend on assumptions that the chapters must make explicit, and some entries are statements whose exact form is still to be fixed. Each entry carries a family tag and a status:

Elementary
The statement is fixed and a short proof is indicated.

Hypotheses to fix
The statement is fixed in outline but the hypotheses must be pinned down in the chapter that proves it.

Specification
The result is described but the precise statement is not yet settled.

The six families are REP (representation and economic knowledge), VIA (viability and comparative statics), PRD (production, deterioration, irreversible investment), INS (institutions, distribution, dependence), CPT (competence, automation, corrective control), and CMP (composition and mathematical interpretation). Identifiers P01 to P29 are stable and are cited by the chapters; the family tag organizes Part IX.

Dependency structure

P01 to P03 are the foundation. P04 to P10, together with P23 and P25, form the theory of sustainable commitments. P11 to P17, together with P24, form the theory of economic representation and control. P26 to P29 add the economic substance for investment, institutions, and compensation. P18 to P22 are the mathematical reinterpretations, and P27 organizes composition across all three. The housing-maintenance example recurs across every group.

Representation and economic knowledge

P01. History sufficiency and continuation equivalence. [REP]

Hypotheses. For an observed history h let $L(h)$ be the set of future action-and-disturbance sequences that satisfy the specified obligations. A summary q is *exact for admissibility* if, for every history h and every future sequence u, whether $u \in L(h)$ is determined by $q(h)$ and u.

Statement. Define $h \sim h'$ if and only if $L(h) = L(h')$. Any exact summary satisfies $q(h) = q(h') \Rightarrow h \sim h'$, so $\sim$ is the coarsest exact summary. If the continuation language has the residual property $L(he) = e^{-1}L(h)$ for each event e, then $\sim$ is a right congruence: $h \sim h' \Rightarrow he \sim h'e$, and the classes support well-defined updates.

Consequence. Debt, deferred maintenance, accumulated exposure, and contractual commitments cannot be dropped merely because current physical holdings match. The result concerns exact admissibility recognition. Predictive sufficiency for probabilities or welfare needs further conditions.

Status. Elementary. Congruence requires the residual property.

P11. Accounting sufficiency requires factorization. [REP]

Statement. Let $F : S \to Y$ be an accounting representation and $P : S \to \{0, 1\}$ an economic property. The property is determined exactly by the accounts if and only if $F(s) = F(s') \Rightarrow P(s) = P(s')$, equivalently $P = \bar{P} \circ F$ on the image of F for some $\bar{P} : F(S) \to \{0, 1\}$.

Relation to P01. P01 and P11 are one factorization lemma. P01 is its dynamic version, where the representation must in addition be a congruence for event updates. Minimal sufficient representation is the quotient by the kernel of the property.

Consequence. Equal reported profit can coexist with different answers about maintenance adequacy, restoration exposure, or viable operation, whenever F identifies states that differ on those properties.

Status. Elementary.

P13. Scalar representation fails for genuine incomparability. [REP]

Statement. (a) If a partial order $\preceq$ on continuation profiles contains incomparable x, y, no real-valued v satisfies $x \preceq y \Leftrightarrow v(x) \leq v(y)$ for all pairs, since $\mathbb{R}$ is totally ordered. (b) Positive companion: every partial order admits a multi-utility representation, a family (v_i) of monotone real functions with $x \preceq y \Leftrightarrow v_i(x) \leq v_i(y)$ for all i. For a finite order the minimal number of functions equals the order dimension (Dushnik and Miller [8]).

Consequence. Useful prices, utilities, and monotone indicators are not ruled out. What is ruled out is an exact, order-reflecting scalar representation of all distinctions.

Status. (a) elementary; (b) hypotheses to fix, in particular the infinite case.

P15. Proxy rewards underdetermine substantive contribution. [REP]

Statement. Let $m(a)$ measure rewarded activity and $V(a)$ the substantive contribution sought. If $m(a) = m(b)$ and $V(a) \neq V(b)$, a reward depending only on m cannot distinguish a from b. If $m(a) > m(b)$ and $V(a) < V(b)$, a strictly increasing function of m ranks them opposite to V.

Consequence. This is a limit on information, not a theory of behaviour. Behaviour requires the participants' costs, available actions, and preferences to be specified separately.

Status. Elementary.

P17. Verification transfers only through an adequate interpretation. [REP]

Hypotheses. Let M be a formal model, E the empirical system, and R a relation connecting their states. Assume soundness of overapproximation: every relevant empirical execution is R-related to a model execution, and R-related states satisfy that model safety implies empirical safety.

Statement. A verified universal safety property of M transfers to E under these hypotheses.

Consequence. Passing a check establishes what its specification and interpretation support. The connection to the economic situation is an additional proof obligation. Liveness and probabilistic claims need stronger relations.

Status. Hypotheses to fix.

P24. Scarcity prices as dual variables. [REP]

Hypotheses. A convex allocation problem with a capacity constraint $\sum_j d_j \leq c$, Slater's condition, and attainment, so that strong duality holds and a multiplier exists.

Statement. There is a multiplier $\lambda^* \geq 0$ supporting the optimal allocation, with complementary slackness $\lambda^*(c - \sum_j d_j) = 0$. If in addition the objective and constraints decompose across participants (separability) and each participant's choice rule is the individual Lagrangian problem at λ^*, then the multiplier is a decentralized price. A multiplier alone does not establish a market implementation.

Consequence. Supporting prices, equilibrium prices, sensitivity derivatives, and sufficient statistics are distinct notions, and uniqueness, decentralizability, and informational sufficiency are separate claims. By P11 a price is a sufficient statistic for a given decision only if that decision factors through it.

Status. Hypotheses to fix.

Viability and comparative statics

P03. Robust viability as a greatest fixed point. [VIA]

Statement. Define $\Phi(X) = \{s \in G : \exists a \in A(s)\ \forall w \in W(s,a),\ T(s,a,w) \in X\}$. The operator Φ is monotone on subsets of S, so by Knaster and Tarski [17] it has a greatest

fixed point $K = \bigcup\{X \subseteq G : X \subseteq \Phi(X)\}$. With $A(s)$ and $W(s,a)$ nonempty and complete observation, K is exactly the set of states from which a policy keeps the system in G forever against all allowed disturbances, and a state-feedback policy suffices.
Convergence. The iteration $K_0 = G$, $K_{n+1} = \Phi(K_n)$ is decreasing. It stabilizes at K in at most $|S|$ steps when S is finite. In general the iteration may need transfinite stages, and the companion code exhibits an infinite chain where each K_n is nonempty but $K = \varnothing$. Termination claims are therefore stated only under separately stated hypotheses. See Aubin [3] for continuous-time viability theory.
Consequence. Present acceptability and sustainable acceptability are different properties.
Status. Elementary for the fixed point and the witnessing action. Convergence needs finiteness.

P23. Comparative statics of viability. [VIA]

Hypotheses. Systems compared on a common state space.
Statement. The kernel K is monotone increasing in the acceptable set G, increasing in each action set $A(s)$, and decreasing in each disturbance set $W(s,a)$, all else equal. The proof is monotonicity of Φ in each parameter, hence of its greatest fixed point.
Consequence. Added duty, added capacity, and added exposure are treated in one statement. P04 and P14 are corollaries.
Status. Elementary.

P04. Obligation tightening cannot enlarge viability. [VIA]

Statement. On a common state space with identical dynamics, actions, and disturbances, $G_2 \subseteq G_1$ implies $K(G_2) \subseteq K(G_1)$. If a new obligation enlarges the state, compare through an explicit simulation relation $\rho \subseteq S_2 \times S_1$ under which ρ-related states respect acceptability and every action and disturbance of the tighter system is matched in the weaker one.
Consequence. Tightening can leave the kernel unchanged, so an added duty does not by itself require extra resources or changed technology. The requirement arises only when the new constraint removes previously viable states or policies, that is, when $K(G_2) \subsetneq K(G_1)$. Preserving the old kernel then requires altering resources, permissions, technology, or disturbance exposure.
Status. Elementary on a common state space. Hypotheses to fix for the simulation version.

P05. Local acceptability does not imply viable continuation. [VIA]

Statement. There exist systems with $s_0 \in K$, $s_1 \in G$, $s_1 \notin K$, and an available action from s_0 to s_1. Three states suffice: from s_0 one action stays at s_0 and another enters s_1, and from s_1 every action enters an unacceptable state $b \notin G$.
Consequence. A current-state check can pass while continued feasibility is destroyed. This supports the trajectory-level analysis of maintenance cuts, cumulative dependence, and repeated interventions.

Status. Elementary. Reported check in the companion (Appendix B).

P06. Safe degradation requires successor viability. [VIA]

Statement. Let $A_K(s) = \{a \in A(s) : T(s, a, w) \in K \text{ for every allowed } w\}$. For $s \in K$, a degradation action preserves the possibility of satisfying the ongoing regime exactly when it lies in $A_K(s)$, and $K = \{s \in G : A_K(s) \neq \varnothing\}$. If degradation creates a restoration duty, that duty must be included in the successor state before A_K is evaluated.
Consequence. Temporary suspension cannot be evaluated apart from the obligation it leaves behind.
Status. Elementary, from P03.

P07. A constructive certificate for compensating continuation. [VIA]

Hypotheses. Let $R \subseteq G$ be the states in which a restoration obligation is discharged. Let $D \subseteq G$ carry a rank $r : D \to \mathbb{N}$ with $r(s) = 0$ exactly on R, and let a policy be such that every allowed successor remains in D and strictly decreases r whenever $r > 0$.
Statement. From $s_0 \in D$ restoration occurs within $r(s_0)$ steps while remaining acceptable. Induction on r.
Consequence. Reserved resources and permissions must support a policy whose progress survives the modelled disturbances.
Status. Elementary.

P14. Corrective control has weak option value. [VIA]

Statement. If $\Pi_0 \subseteq \Pi_1$ are policy sets for identical environments and criteria, $\sup_{\Pi_1} J \geq \sup_{\Pi_0} J$. Strict improvement requires a circumstance in which the added action helps. For viability the same statement is the action-set case of P23.
Consequence. Corrective capacity has potential value, but its acquisition is a substantive investment problem. Training, monitoring, reserves, and intervention infrastructure must be priced explicitly.
Status. Elementary.

P25. Almost-sure and risk-constrained viability. [VIA]

Settled statement (countable case). Robust viability over disturbance supports is not in general equivalent to almost-sure viability. In a finite or countable model, requiring every positive-probability successor to stay in X gives the almost-sure predecessor condition. For general kernels, the operator to be studied is $\Phi_{\text{as}}(X) = G \cap \{s : \exists a \in A(s),\ \mathbb{P}(X \mid s, a) = 1\}$.
Hypotheses to fix. Identifying the greatest fixed point of Φ_{as} with an implementable policy needs measurability and selection assumptions. Moving to universally measurable or analytic sets does not by itself solve the problem, since those classes are not closed under all the operations and transfinite iterations involved (Bertsekas and Shreve [4]). The measurable domain and the policy-selection theorem must be fixed first. A support formulation holds

for closed X, where full measure implies containment of the support. Risk-constrained viability needs an augmented state carrying a remaining risk budget and a justified rule for allocating conditional risk across successors.
Status. Countable case: elementary. General-state version: research specification until the measurable domain and selection theorem are fixed.

Production, deterioration, and irreversible investment

P12. Financial improvement can accompany capability depletion. [PRD]
Statement. Let surplus be $\pi = R - C - M$ with maintenance expenditure M, and capacity evolve by $k_{t+1} = k_t - d(k_t) + g(M_t)$. (a) If a reduction $\delta > 0$ in maintenance leaves revenue and other costs unchanged, current surplus rises by δ, and if g is strictly increasing over the relevant interval, successor capacity falls relative to the maintained baseline. (b) Threshold: derive the viability boundary $k^*(\bar{M})$ for a maintenance budget $\bar{M}$, and state conditions under which it jumps or has a kink.
Consequence. Nonlinear deterioration alone does not guarantee a discontinuity or a kink. The threshold result must say what does.
Status. (a) elementary. (b) specification.

P26. When waiting improves attainable value. [PRD]
Elementary two-period form. Investment of cost I yields a payoff X with mean μ, revealed after one period. Acting now yields $V_{\text{act}} = (\mu - I) + r$, where r is the first-period flow return forgone by waiting. Waiting yields $V_{\text{wait}} = \beta\,\mathbb{E}[\mathbf{1}_{\text{surv}}(X - I)^+]$ with discount β, where $\mathbf{1}_{\text{surv}}$ indicates that the opportunity survives. If survival is independent of the revealed payoff with probability q, this equals $\beta q\,\mathbb{E}[(X - I)^+]$. Otherwise it does not, and the dependence must be modelled. Waiting is better exactly when $V_{\text{wait}} > V_{\text{act}}$. By Jensen, $\mathbb{E}[(X - I)^+] \geq (\mu - I)^+$, so the option value of information is nonnegative before discounting, deterioration, and expiry. See Dixit and Pindyck [7].
General statement. Compare the value of information acquired through delay with deterioration, forgone returns, and expiring opportunities, treating action and inaction symmetrically as potentially irreversible.
Status. Two-period form elementary. General form specification.

Institutions, distribution, and dependence

P02. Establishment precedes selection. [INS]
Statement. Let D be an opportunity domain, $E(e, D)$ the domain established by an acquisition action e, and $C(D)$ the outcomes obtainable by selection inside D. If $x \notin C(D)$ and

$x \in C(E(e, D))$, selection within D cannot implement x whereas establishment followed by selection can. Budgeted version: assume a zero-cost identity establishment action (or an explicit permission to skip establishment). Then with budget b, outcomes under establish-then-select policies contain those under selection only, and the inclusion is strict only if an affordable establishment action makes an otherwise inaccessible outcome reachable.
Consequence. Expenditure on access, infrastructure, learning, or recognition can change the choice problem itself.
Status. Elementary.

P08. Nominal exit and executable exit are distinct. [INS]

Statement. Let X be an exit target and B the states satisfying the safety and obligation constraints during departure. Define $\text{Reach}_0 = X$ and $\text{Reach}_{k+1} = \text{Reach}_k \cup \{s \in B : \exists a \ \forall w, \ T(s, a, w) \in \text{Reach}_k\}$. An exit is executable within horizon H exactly when $s \in \text{Reach}_H$. A legal permission to exit does not imply membership. A permitted departure requiring a fee beyond the person's accessible resources is a counterexample.
Consequence. Exit capacity is an operational variable and its distribution can be studied independently of whether contracts contain an exit clause.
Status. Elementary.

P28. Surplus gain with a shrunken safe exit set. [INS]

Statement. Characterize when an intervention raises aggregate surplus while reducing a specified participant's executable exit set, building on P08, and relate this to bargaining, cumulative contractual dependence, and portability.
Status. Proved in the hand-calculable housing model (Theorem H1 in the housing case). General statement: specification.

P29. When compensation restores capability. [INS]

Statement. Identify conditions under which a transfer restores the lost capability rather than merely increasing monetary holdings: the compensating bundle must lie in the reachability preimage of the lost capability under prices and access actually available, using P08 and P07.
Status. Proved in the housing model (Theorem H2 in the housing case): restoration requires availability, timing, and sufficiency. General statement: specification.

Competence, automation, and corrective control

P16. Safe aspect relegation requires a detection-and-return margin. [CPT]

Statement. Let an automated operating region lie at distance d from a failure boundary, with adverse motion at rate at most v, detection and return to explicit control within time τ,

and intervention able to arrest deterioration before the boundary. Then $d > v\tau$ is sufficient, plus allowances for observation error and stopping distance. Endogenous version: response time depends on retained competence, $\tau_t = \tau(c_t)$, and $c_{t+1} = f(c_t, u_t, \ell_t)$ with practice u_t and reliance on relegated operation ℓ_t. Under monotonicity assumptions (τ decreasing in c, f increasing in u and decreasing in ℓ), the condition is $d_t > v_t\,\tau(c_t) + \text{allowances}$.
Economic result. The minimal practice expenditure u_t^* that preserves the margin, and the circumstances in which apparent automation savings are consumed by it.
Status. Fixed-τ form elementary. Endogenous form hypotheses to fix.

Composition and mathematical interpretation

P09. Local feasibility is insufficient under shared scarcity. [CMP]

Statement. Two commitments each needing one unit of a resource at the same time, with one unit available, are each feasible alone and jointly infeasible. For divisible simultaneous demands d_j on capacity c, $d_j \le c$ for every j does not imply $\sum_j d_j \le c$, and $\sum_j d_j \le c$ is necessary and sufficient for joint feasibility in the static divisible case.
Consequence. This is the obstruction every later compositional guarantee must address. Departments, firms, and agencies can each hold feasible plans that cannot be executed together.
Status. Elementary. Reported check in the companion (Appendix B).

P10. Independent viability composes under explicit independence. [CMP]

Hypotheses. Two systems with product state space, independent actions, product disturbance sets, factorized dynamics, nonempty action and disturbance sets, and acceptable set $G_1 \times G_2$.
Statement. The robust kernel is $K_{12} = K_1 \times K_2$. For $\supseteq$, combine the witnessing policies. For $\subseteq$, a joint action may depend on both components, so the projection is not immediate. Fix an allowed disturbance selection for system 2 and simulate system 2 inside a policy for system 1. The induced policy depends only on system 1's history and keeps it in G_1 against all of its disturbances. This is where nonemptiness and the policy class matter.
Consequence. Shared budgets, coupled timing, or cross-system effects violate the premises. The theorem states exactly what must be established before local guarantees assemble without further analysis.
Status. Hypotheses to fix. Reported check in the companion, with and without coupling.

P27. Circular assume-guarantee composition. [CMP]

Statement. Components carry assumption-guarantee pairs (A_i, G_i) on safety properties, with shared capacity appearing in the interface as a time-indexed assumption on the combined draw. If each guarantee at step $n+1$ depends only on assumptions through step n,

the circular argument is well founded by induction on n, and the conjunction of guarantees holds for the composition (Abadi and Lamport [1]).
Consequence. The temporal discipline is what prevents circular assumptions from justifying themselves. It restricts the argument to safety properties.
Status. Specification. The inductive form is the target statement.

P18. Economic processes form a category when interfaces compose. [CMP]

Hypotheses. Let $\mathcal{P}$ be the full path category: objects are interfaces, morphisms are all histories between them (admissible or not), composition is concatenation. Let Adm be a set of morphisms of $\mathcal{P}$, the admissible histories.
Statement. For parallel morphisms $f, g : A \to B$ define $f \approx g$ if for every pair of composable contexts $u : C \to A$ and $v : B \to D$, $v \circ f \circ u \in \text{Adm}$ if and only if $v \circ g \circ u \in \text{Adm}$. Only parallel morphisms are compared, so composition preserves sources and targets. Then $\approx$ is a congruence for composition, $\mathcal{P}/\approx$ is a category (the syntactic category of Adm) with identities the classes of empty histories, and Adm is a union of $\approx$-classes, so it is recognized within the quotient. The admissible histories themselves need not form a subcategory: they need not be closed under composition, as P05 and P09 show, and constructing the quotient of the full path category avoids reintroducing that closure problem. A monoidal structure needs a parallel composition satisfying the interchange law, which fails under shared resources unless the interface records them (P09, P10). See Mac Lane [12].
Consequence. One-sided continuation equivalence from P01 gives only a right congruence. Quotienting morphisms needs stability under both preceding and following composition, so the categorical chapter begins with this syntactic congruence, or with the transition category of the minimized continuation system.
Status. Hypotheses to fix.

P19. Categorical accounting preserves specified composition laws. [CMP]

Statement. An accounting interpretation is a functor when $F(g \circ f) = F(g) \circ F(f)$ and $F(\text{id}_s) = \text{id}_{F(s)}$. For a chosen accounting system these equations follow from its posting rules. Its limitations are then analyzed through P11: a composition-preserving representation can still forget distinctions needed to assess continuation.
Consequence. Coherent bookkeeping and sufficient economic description are different properties.
Status. Elementary for a given posting system.

P20. Typed execution: preservation and progress. [CMP]

Statement. Let $\Gamma \vdash a : s \Rightarrow s'$ require the stated permissions, account for resource use, and update obligations. If every primitive rule is sound for the transition semantics, sequential composition yields sound judgments by induction on derivations, and well-typed execution maintains a specified invariant (*preservation*). *Progress*: a well-typed commitment

is executable, conditional on available resources, enabled counterparties, and any environmental assumptions needed for execution. Linear or affine disciplines prevent duplication of exclusive resources when the semantics and trusted boundary enforce them.
Consequence. The gap between formal inhabitation and executability is exactly where progress can fail. The theorem does not establish the truth of sensor readings, the legitimacy of obligations, or the completeness of the model.
Status. Preservation elementary. Progress hypotheses to fix.

P21. Compatible local records glue exactly under the sheaf condition. [CMP]
Statement. Let economic contexts form a site and let $\mathcal{F}$ assign records with coherent restriction maps. If $\mathcal{F}$ is a sheaf, every matching family on a cover has a unique global section.
Consequence. The substantive work is proving that a particular record system satisfies the condition. Feasible allocations can fail to form a sheaf when local contexts omit globally shared capacity, as in P09. Gluing records and establishing executable coordination are separate tasks unless the construction links them.
Status. Hypotheses to fix, per record system.

P22. A cohomological obstruction in an affine coordination model. [CMP]
Statement. If compatible configurations form a torsor under an abelian sheaf $\mathcal{A}$, local choices give transition differences on overlaps, forming a degree-one cocycle. A global configuration exists exactly when the torsor class vanishes in $H^1(X, \mathcal{A})$, under the standard classification hypotheses (Mac Lane and Moerdijk [13]).
Worked case, frictionless. On a graph give each oriented edge a log exchange rate c_{ij} with $c_{ji} = -c_{ij}$. A global log-price potential p with $c_{ij} = p_j - p_i$ exists exactly when the sum around every cycle is zero, that is, when the class $[c] \in H^1(\Gamma; \mathbb{R})$ vanishes. A nonzero cycle sum gives an arbitrage direction.
Worked case, frictions. With bid–ask spreads, let u_{ij} be an upper bound on the log rate from i to j. A potential with $p_j - p_i \le u_{ij}$ exists if and only if there is no negative cycle (shortest-path duality, Bellman–Ford), and the vanishing of $[c]$ is no longer required. The frictionless case is the limit $u_{ji} = -u_{ij}$, where the two inequalities force equality.
Consequence. The example gives a genuine obstruction theory for the model. Broader institutional conflicts need separate constructions and must not be described as cohomological without one.
Status. Graph case elementary, with a reported check in the companion. General sheaf case specification.

Where each result appears

Chapter assignments are provisional. For each result, the chapters are grouped by where it is stated early (Parts I to VIII), where it is stated rigorously (Part IX), and where it is

reinterpreted (Parts X to XII). A dash means the band has no assigned chapter.

Result	Family	Early (I–VIII)	Rigorous (IX)	Reinterpreted (X–XII)
P01	REP	2	41	46
P02	INS	4, 39	41	51
P03	VIA	13	43	54
P04	VIA	13	43	50
P05	VIA	15	43	46
P06	VIA	12	43	54
P07	VIA	14	43	53
P08	INS	3, 29	43	53
P09	CMP	31	44	47, 58
P10	CMP	31	44	47
P11	REP	20, 21	41	49
P12	PRD	8, 18	45	49
P13	REP	16	42	48
P14	VIA	15	43	54
P15	REP	35	45	55
P16	CPT	23, 24	45	54
P17	REP	25	45	55
P18	CMP	31	44	46, 47, 60
P19	CMP	20	41	49
P20	CMP	12	–	51, 52, 53, 54, 60
P21	CMP	22	–	56, 57, 58, 60
P22	CMP	17	–	59, 60
P23	VIA	13	43	50
P24	REP	17	44	48
P25	VIA	13	43	–
P26	PRD	9, 40	43	–
P27	CMP	32	44	50
P28	INS	29	45	53
P29	INS	19	45	49

Running example

A housing-maintenance system carries all six families. Deterioration establishes the dynamics. Tenancy and repair duties establish obligations. Contractor availability determines

progress. Budgets introduce scarcity prices. Records test representation sufficiency. Shared crews expose the failure of local composition. The finite-state companion reports checks of the viability and independence results and deliberately reproduces counterexamples when their hypotheses are removed, so that the proof programme is a working specification and not a parallel collection of formal propositions.

Part I

Economic Activity and the Establishment of Futures

Part I fixes what an economy maintains and introduces the objects the rest of the book relies on: states, histories, continuations, and the distinction between what is possible, accessible, permitted, and acceptable.

HOUSING THREAD. Define the building, its tenants, and the repair crew, and list the states and histories that matter: rent arrears, deferred work, and notices served.

Chapter 1

What an Economy Maintains

Introduces an economy as an organization of provisioning, transformation, and maintenance through time, using a household, a construction firm, a hospital, and a watershed to show the range of relevant processes. Separates current output from the reproduction of the conditions that make output possible, and shows how one transaction can generate income while exhausting productive capacity elsewhere.

Chapter scaffold: motivating situation; definitions and model; developed argument; worked example; results with proofs where warranted; empirical questions; exercises.

Chapter 2

States, Histories, and Continuations

Introduces economic states, histories, actions, and transition relations. States when a present-state description is sufficient and when accumulated obligations, deterioration, learning, or institutional commitments require history to remain explicit. The basic object is a set of possible trajectories, not a single forecast. The first theorem characterizes sufficient state augmentation: a compressed history is adequate when histories with the same summary have equivalent consequences for the future questions at issue.

$$x_{t+1} \in T(x_t, h_t, a_t), \qquad h_{t+1} = U(h_t, x_t, a_t, x_{t+1}).$$

Proof programme. P01.

Chapter scaffold: motivating situation; definitions and model; developed argument; worked example; results with proofs where warranted; empirical questions; exercises.

Chapter 3

Feasibility, Accessibility, and Admissibility

Separates physically possible actions, practically accessible actions, institutionally permitted actions, and actions consistent with stated obligations. Examines why formal eligibility does not establish effective access and why an available choice need not be an acceptable continuation. Examples are housing, employment, medical treatment, and access to credit.

Proof programme. P08.

Chapter scaffold: motivating situation; definitions and model; developed argument; worked example; results with proofs where warranted; empirical questions; exercises.

Chapter 4

Establishing the Domain of Choice

Examines the work required to make alternatives available before selection can occur. Search, education, infrastructure, legal recognition, invention, and interface design become explicit economic activities. A two-stage model distinguishes expenditures that establish an opportunity domain from expenditures that select within it.

Proof programme. P02.

Chapter scaffold: motivating situation; definitions and model; developed argument; worked example; results with proofs where warranted; empirical questions; exercises.

Chapter 5

Welfare Without Counting Options

Addresses the normative foundations directly. More possible trajectories do not automatically make a better life, and constraints can protect some people while excluding others. Develops plural assessments involving substantive achievement, security, burden, access, and corrective freedom, and makes disagreements about those assessments explicit.

Chapter scaffold: motivating situation; definitions and model; developed argument; worked example; results with proofs where warranted; empirical questions; exercises.

Part II

Production, Maintenance, and Irreversibility

Part II treats production as the transformation of conditions, capital as maintained capability, and irreversibility as a symmetric property of action and inaction.

HOUSING THREAD. Roof and plumbing deteriorate; producing a habitable unit transforms the building's conditions; the cost of deferred maintenance is traced forward.

Chapter 6

Production as Transformation of Conditions

Represents production through its outputs, consumed inputs, residuals, new obligations, and changes to future productive capacity. Compares a production function with a richer transition description and shows which questions each representation can answer.

Chapter scaffold: motivating situation; definitions and model; developed argument; worked example; results with proofs where warranted; empirical questions; exercises.

Chapter 7

Capital as Maintained Capability

Examines machines, buildings, knowledge, networks, and institutional arrangements as capabilities that require continued support. Capital valuation depends on complementary inputs, repairability, skills, and access, not on possession alone. A worked example compares two machines with identical output but different maintenance dependencies.

Chapter scaffold: motivating situation; definitions and model; developed argument; worked example; results with proofs where warranted; empirical questions; exercises.

Chapter 8

Maintenance and the Reproduction of Capacity

Develops maintenance schedules, deterioration processes, deferred work, and critical thresholds. A formal model shows how short-run expenditure reductions can reduce long-run feasible output when deterioration is nonlinear. Preventive maintenance is distinguished from restoration after failure.

Proof programme. P12.

Draft note. Correction to carry over from earlier manuscripts: a nonpositive long-run average net accumulation rate does not ensure bounded debt, since $D(t) = \sqrt{1+t} - 1$ is unbounded with average rate tending to zero. A bounded-debt theorem needs $\sup_{t \geq 0} \int_0^t (\rho_M - \rho_R)\, du < \infty$, or a reflected-backlog formulation.

Chapter scaffold: motivating situation; definitions and model; developed argument; worked example; results with proofs where warranted; empirical questions; exercises.

Chapter 9

Irreversible Decisions and the Value of Waiting

Connects continuation economics to established work on irreversible investment and option value. Examines when waiting preserves useful information and when delay itself destroys opportunities. The result is a symmetric treatment of action and inaction as potentially irreversible.

Proof programme. P26.

Chapter scaffold: motivating situation; definitions and model; developed argument; worked example; results with proofs where warranted; empirical questions; exercises.

Chapter 10

Residuals, Waste, and the Afterlives of Production

Investigates demolition, abandoned infrastructure, pollution, obsolete software, and administrative residues. Asks what persists after an activity's advertised purpose has ended, and extends responsibility to the disposition of those residuals.

Chapter scaffold: motivating situation; definitions and model; developed argument; worked example; results with proofs where warranted; empirical questions; exercises.

Part III

Obligations, Degradation, and Recovery

Part III represents obligations through time and introduces viability kernels, safe degradation, compensating continuation, and corrective capacity.

HOUSING THREAD. Tenancy and repair duties become obligations, an unheated unit may not be let, and viability is studied under a maintenance budget.

Chapter 11

Obligations Through Time

Represents contracts, promises, legal duties, maintenance commitments, and informal responsibilities as obligations with bearers, beneficiaries, triggers, deadlines, and discharge conditions. Explains why obligations cannot generally be inferred from a balance sheet or a snapshot of resources.

Chapter scaffold: motivating situation; definitions and model; developed argument; worked example; results with proofs where warranted; empirical questions; exercises.

Chapter 12

Required Functions and Optional Capabilities

Distinguishes functions that must continue from capabilities that can be reduced or suspended, and examines how these classifications are established, challenged, and revised. A public-service example shows that declaring a function optional is itself a consequential institutional decision.

Proof programme. P06, P20.

Chapter scaffold: motivating situation; definitions and model; developed argument; worked example; results with proofs where warranted; empirical questions; exercises.

Chapter 13

Viability Under Scarcity

Introduces viability kernels and constrained reachability, and distinguishes a desirable target from the set of states from which obligations can still be fulfilled. Uncertainty is treated separately for deterministic, robust, and probabilistic models. Comparative statics for added duties, capacity, and exposure are stated together.

$K = \{x : \exists \pi \text{ such that all trajectories permitted by the model under } \pi \text{ satisfy the continuation constraints}\}.$

Proof programme. P03, P04, P23, P25.

Chapter scaffold: motivating situation; definitions and model; developed argument; worked example; results with proofs where warranted; empirical questions; exercises.

Chapter 14

Compensating Continuation

Links suspension to restoration obligations, deadlines, and reserved resources. Distinguishes credible compensation from promises that depend on capabilities already being consumed. Worked cases are deferred building repairs, reduced hospital services, and temporary infrastructure closures.

Proof programme. P07.

Chapter scaffold: motivating situation; definitions and model; developed argument; worked example; results with proofs where warranted; empirical questions; exercises.

Chapter 15

Corrective Capacity and Recoverability

Defines the resources, permissions, knowledge, and time required to correct a course of action. Examines systems that remain operational while losing the ability to recover from foreseeable disturbances. A theorem gives conditions under which locally admissible steps leave the admissible region over a longer horizon.

Proof programme. P05, P14.

Chapter scaffold: motivating situation; definitions and model; developed argument; worked example; results with proofs where warranted; empirical questions; exercises.

Part IV

Value, Prices, Accounting, and Debt

Part IV compares valuations, prices, profit, debt, and accounts as representations, and asks what each preserves of the underlying continuation structure.

HOUSING THREAD. Rent, reserves, and the landlord's reported profit are set beside the maintenance backlog, which no conventional account records as a liability.

Chapter 16

Valuing Feasible Futures

Develops valuations over trajectories, capability bundles, and contingent opportunities. Monetary valuation is one representation among several. Identifies circumstances in which scalar representations exist and circumstances in which relevant comparisons remain partial.

Proof programme. P13.

Draft note. Candidate import: the identity $D_{\mathrm{KL}}(P(\cdot \mid B) \,\|\, P) = -\log P(B)$ for $0 < P(B) \leq 1$ as a measure of probabilistic restriction. It does not by itself give a monetary cost, a physical dissipation, or a conservation law.

Chapter scaffold: motivating situation; definitions and model; developed argument; worked example; results with proofs where warranted; empirical questions; exercises.

Chapter 17

Prices as Compressed Coordination Signals

Studies prices as economical representations that permit decentralized coordination while omitting much of the underlying causal structure. Asks which omissions are tolerable for a given decision and which require supplementary records, standards, or institutions. Scarcity prices appear as dual variables, and an exchange-rate graph gives the first obstruction example.

Proof programme. P22, P24.

Draft note. Candidate import: allocation of finite maintenance capacity, $\max_{r_i \geq 0} \sum_i V_i(r_i)$ subject to $\sum_i r_i \leq B$, with the budget an explicit assumption and the shadow price as dual variable. No universal conservation of ambiguity is assumed.

Chapter scaffold: motivating situation; definitions and model; developed argument; worked example; results with proofs where warranted; empirical questions; exercises.

Chapter 18

Profit and the Location of Costs

Decomposes financial profitability into revenues, recognized costs, transfers of burden, and changes in maintained capacity. A worked enterprise model distinguishes profit supported by reproducible productivity from profit supported by deferred replacement or unpriced damage.

Proof programme. P12.

Chapter scaffold: motivating situation; definitions and model; developed argument; worked example; results with proofs where warranted; empirical questions; exercises.

Chapter 19

Monetary Debt and Restoration Obligations

Distinguishes financial liabilities from duties to repair ecological, infrastructural, or social damage. Explores when obligations can be transferred or monetized and when substitution fails because the relevant beneficiary, place, or lost capability matters.

Proof programme. P29.

Chapter scaffold: motivating situation; definitions and model; developed argument; worked example; results with proofs where warranted; empirical questions; exercises.

Chapter 20

Accounting for Continuation

Proposes supplementary accounts for maintenance backlogs, recovery reserves, critical dependencies, and outstanding disposition duties. These accompany conventional accounts without presuming that every dimension can be aggregated into one number. The practical aim is auditable visibility.

Proof programme. P11, P19.

Draft note. Candidate case (documentation-level only): a two-ledger design that keeps official spending commitments separate from community observations and reports their divergence instead of averaging them. Distinguish an obligation, a reported action, evidence of performance, and discharge of the obligation as four different institutional events.

Chapter scaffold: motivating situation; definitions and model; developed argument; worked example; results with proofs where warranted; empirical questions; exercises.

Part V

Knowledge, Attention, and Productive Competence

Part V studies representations, memory, skill, automation, and verification as supports for, or threats to, competent continued action.

HOUSING THREAD. Inspection records, contractor knowledge, and automated scheduling are examined, along with what a passed inspection establishes.

Chapter 21

Economic Representations and Semantic Compression

Examines the representations used in planning, forecasting, classification, and coordination. A useful representation preserves distinctions relevant to action while compressing others. Defines task-relative representational adequacy.

Proof programme. P11.

Chapter scaffold: motivating situation; definitions and model; developed argument; worked example; results with proofs where warranted; empirical questions; exercises.

Chapter 22

Memory as Organizational Capital

Examines records, provenance, routines, and institutional memory as supports for continued competent action. Distinguishes retained information from retrievable and usable knowledge, and investigates when forgetting reduces costs without erasing obligations.

Proof programme. P21.

Draft note. Candidate import: separate replay integrity from historical authority. A self-consistent checkpoint is not an independently trusted history, and branch selection is not evidence that the chosen branch is true. An institution can keep reproducible accounts without establishing that inputs are truthful or commitments legitimate.

Chapter scaffold: motivating situation; definitions and model; developed argument; worked example; results with proofs where warranted; empirical questions; exercises.

Chapter 23

Aspect Relegation and the Economics of Skill

Models practice and routine as changes in the attention required to perform a task adequately. Relegation frees resources while creating dependence on stable conditions. Develops the economic significance of recognizing when a task must return to explicit supervision.

Proof programme. P16.

Draft note. Candidate import: distinguish expenditure from damage to regenerative capacity, and allocate finite repair capacity through explicit response functions.

Chapter scaffold: motivating situation; definitions and model; developed argument; worked example; results with proofs where warranted; empirical questions; exercises.

Chapter 24

Automation and the Preservation of Intervention

Assesses automation through output, error, oversight burden, retained competence, and recovery time. A formal example shows why removing human involvement can lower immediate costs while raising the cost of rare failures, with response time made endogenous to retained competence.

Proof programme. P16.

Draft note. Candidate case (documentation-level only): an agent-automation tool. Isolated working copies are not an operating-system security boundary, and a command filter or stop control cannot recover effects that have already occurred. The economic question is how much review, intervention time, and recoverability the automation actually preserves. Exclude any zero-risk claim.

Chapter scaffold: motivating situation; definitions and model; developed argument; worked example; results with proofs where warranted; empirical questions; exercises.

Chapter 25

Verification, Proxies, and Institutional Confidence

Distinguishes what an audit, benchmark, certification, or formal proof actually establishes. Examines the economic effects of treating a passed check as broader evidence than its specification warrants.

Proof programme. P17.

Draft note. Candidate counterexample: a ledger verifier that checks seals against stored roots without recomputing them from contents, and checks debits one at a time without aggregating repeated postings, so that individually permissible debits jointly exceed the stated balance restriction. A check can succeed while omitting the relation a reader assumes it verifies.

Chapter scaffold: motivating situation; definitions and model; developed argument; worked example; results with proofs where warranted; empirical questions; exercises.

Part VI

Institutions, Power, and Distribution

Part VI analyzes classifications, property, contracts, exit, and public authority as distinctions that alter which continuations are accessible and to whom.

HOUSING THREAD. Tenure, deposits, notice periods, and a tenant's executable capacity to leave are placed in one model.

Chapter 26

Distinctions That Allocate Futures

Examines property categories, employment classifications, citizenship, credit status, and professional credentials as distinctions that alter accessible continuations. Connects the theory of distinction and continuation to institutional economics.

Chapter scaffold: motivating situation; definitions and model; developed argument; worked example; results with proofs where warranted; empirical questions; exercises.

Chapter 27

Property as Structured Authority

Analyzes property through permissions to use, exclude, modify, transfer, encumber, and abandon. Distinguishes ownership from the practical capacity to exercise those permissions and from the burdens imposed on others.

Chapter scaffold: motivating situation; definitions and model; developed argument; worked example; results with proofs where warranted; empirical questions; exercises.

Chapter 28

Contracts and the Evolution of Dependence

Studies contracts as trajectory-shaping arrangements. Examines how repeated agreements alter bargaining positions, switching costs, and the conditions of later consent, and analyzes cumulative dependence under continuing intervention.

Chapter scaffold: motivating situation; definitions and model; developed argument; worked example; results with proofs where warranted; empirical questions; exercises.

Chapter 29

Distribution of Exit and Refusal

Treats inequality as including unequal capacities to leave employment, housing, platforms, financial relationships, and harmful environments. Distinguishes nominal exit rights from executable exit paths, and states when an intervention can raise surplus while shrinking a participant's safe exit set.

Proof programme. P08, P28.

Chapter scaffold: motivating situation; definitions and model; developed argument; worked example; results with proofs where warranted; empirical questions; exercises.

Chapter 30

Public Authority and the Admissibility of Constraints

Addresses democratic authorization, contestability, rights, and institutional revision. Examines who establishes continuation constraints, how conflicts among them are adjudicated, and what prevents preservation language from legitimating domination.

Draft note. Candidate question: what must happen before a repair obligation may legitimately be discharged? A waiting period is an enforceable procedural condition but elapsed time cannot establish successful repair. Closure should require procedural authorization and specified evidence, with a route for reopening a failed repair. Grouping reports by location suppresses duplicate influence but does not establish independent witnesses, and a minimum cluster size does not by itself protect against all differencing attacks. Present as a proposed institutional design, not evidence of deployment.

Chapter scaffold: motivating situation; definitions and model; developed argument; worked example; results with proofs where warranted; empirical questions; exercises.

Part VII

Markets, Firms, Commons, and Coordination

Part VII asks when individually acceptable commitments compose into a feasible collective arrangement, in markets, firms, platforms, and commons.

Housing thread. A shared crew serves several buildings, listings run through a platform, and a cooperative maintenance fund pools contributions.

Chapter 31

Exchange and Compatible Commitments

Models exchange as a transformation of resources, permissions, and obligations among participants. Identifies conditions under which individually acceptable exchanges compose into a feasible collective arrangement.

Proof programme. P09, P10, P18.

Draft note. Candidate case (documentation-level only): two customers each receive an apparently feasible offer for the last item while one purchase can be fulfilled. Reservations add time: resources are withheld, expire, and return. Authorization and resource commitment must stay consistent under concurrent requests (P09).

Chapter scaffold: motivating situation; definitions and model; developed argument; worked example; results with proofs where warranted; empirical questions; exercises.

Chapter 32

Firms as Organizations of Complementary Continuation

Treats firms as coordinating tasks, assets, knowledge, and promises that cannot be maintained through isolated transactions alone. Boundaries of the firm are examined through coordination costs, authority, dependence, and the distribution of residual responsibilities.

Proof programme. P27.

Chapter scaffold: motivating situation; definitions and model; developed argument; worked example; results with proofs where warranted; empirical questions; exercises.

Chapter 33

Platforms and Infrastructural Dependence

Studies platforms as institutions that establish opportunity domains while controlling access to them. Examines interoperability, portability, unilateral rule changes, and the appropriation of complementary investment.

Draft note. Candidate import: revenue with a participation term, $R(q) = (\alpha q + \beta)M(q)$, so $R'(q) = \alpha M(q) + (\alpha q + \beta)M'(q)$. Revenue can fall as match quality rises if participation falls strongly enough with quality. This is a conditional incentive result only.

Chapter scaffold: motivating situation; definitions and model; developed argument; worked example; results with proofs where warranted; empirical questions; exercises.

Chapter 34

Commons and Shared Maintenance

Analyzes shared resources through contribution rules, monitoring, access, replenishment, and conflict resolution. Distinguishes common ownership from the actual organization of maintenance.

Chapter scaffold: motivating situation; definitions and model; developed argument; worked example; results with proofs where warranted; empirical questions; exercises.

Chapter 35

Contribution, Reward, and Measurement

Uses work-backed currencies, contribution scores, and automated reward systems as worked cases. Shows how reward mechanisms change the activity they measure and why maintenance and preventive work are particularly hard to recognize through simple activity counts.

Proof programme. P15.

Chapter scaffold: motivating situation; definitions and model; developed argument; worked example; results with proofs where warranted; empirical questions; exercises.

Part VIII

Ecology, Care, Technology, and Development

Part VIII applies the framework to ecological reproduction, care, productive infrastructure, development, and coordinated transitions, and closes the economic text.

HOUSING THREAD. An energy retrofit is a transition with a viable path, and damp and mould are ecological conditions of the dwelling.

Chapter 36

Ecological Reproduction and Economic Time

Introduces ecosystems as dynamic conditions of provisioning with thresholds, regeneration rates, and spatial dependencies. Investigates conflicts between financial horizons and ecological restoration times.

Chapter scaffold: motivating situation; definitions and model; developed argument; worked example; results with proofs where warranted; empirical questions; exercises.

Chapter 37

Care and the Maintenance of Persons

Treats care as skilled, materially supported work that sustains people's capacities and relationships. Addresses unpaid labour, dependency, exhaustion, and the danger of valuing persons solely through productive contribution.

Chapter scaffold: motivating situation; definitions and model; developed argument; worked example; results with proofs where warranted; empirical questions; exercises.

Chapter 38

Local AI and the Ownership of Productive Infrastructure

Uses local inference, rented cloud services, persistent agents, and open software as a sustained case study. Compares total costs, supplier dependence, maintenance burdens, portability, and the distribution of control.

Chapter scaffold: motivating situation; definitions and model; developed argument; worked example; results with proofs where warranted; empirical questions; exercises.

Chapter 39

Development as the Establishment of Capabilities

Treats development as building accessible domains of action through education, health, infrastructure, legal security, and collective competence. Examines the tension between increasing output and preserving communities' authority over the conditions of their lives.

Proof programme. P02.

Chapter scaffold: motivating situation; definitions and model; developed argument; worked example; results with proofs where warranted; empirical questions; exercises.

Chapter 40

Crisis, Transition, and Institutional Repair

Culminates the economic text in coordinated transitions involving energy, housing, public finance, and industrial change. Explains why a viable destination is insufficient without a viable path, and examines transition funding, temporary protections, restoration duties, and mechanisms for revising a failing plan.

Proof programme. P26.

Chapter scaffold: motivating situation; definitions and model; developed argument; worked example; results with proofs where warranted; empirical questions; exercises.

Part IX

Mathematical Foundations and Economic Results

Part IX consolidates the formal machinery and states the principal results with hypotheses, organized by theorem family. It is the formal centre of the book.

HOUSING THREAD. The housing model is stated formally, its viability kernel is computed, and the finite-state companion reproduces the results and their counterexamples.

Chapter 41

Transition Systems and History Compression

Consolidates the formal machinery used throughout the book. Establishes sufficient-state conditions, observational equivalence, and history-dependent constraints, with careful treatment of what a chosen state representation forgets. The representation family is stated here as a single factorization lemma with a dynamic congruence version.

Proof programme. P01, P02, P11, P19.

Chapter scaffold: motivating situation; definitions and model; developed argument; worked example; results with proofs where warranted; empirical questions; exercises.

Chapter 42

Orders, Dominance, and Incomparability

Equips continuation profiles with partial orders reflecting specified dimensions of improvement. Establishes conditions for dominance and explains why inclusion, cardinality, and scalar utility are different comparisons. Any impossibility result concerning scalarization states the preservation requirements that generate it, and the multi-utility companion is given.

Proof programme. P13.

Chapter scaffold: motivating situation; definitions and model; developed argument; worked example; results with proofs where warranted; empirical questions; exercises.

Chapter 43

Reachability, Robustness, and Recovery

Develops finite-state algorithms and selected continuous models for viability and corrective reachability, with the fixed-point theorem and comparative statics. Compares worst-case guarantees with probabilistic guarantees and analyzes the effects of model uncertainty, including almost-sure and risk-budget variants. Includes the waiting theorem.

Proof programme. P03, P04, P05, P06, P07, P08, P14, P23, P25, P26.

Chapter scaffold: motivating situation; definitions and model; developed argument; worked example; results with proofs where warranted; empirical questions; exercises.

Chapter 44

Composition and Systemic Failure

Assembles local contracts into networks with shared resources and timing constraints. Counterexamples show how individually feasible commitments become jointly infeasible. Positive results identify sufficient compatibility and resource conditions, including independent composition, scarcity prices, and circular assume-guarantee reasoning.

Proof programme. P09, P10, P18, P24, P27.

Draft note. Candidate case: a transactional outbox records the obligation to send an event in the same transaction as the order, which prevents one class of omission. Without further assumptions it does not guarantee eventual delivery or prevent duplicate effects at the recipient, which separates accounting consistency from operational progress (P20, P27).

Chapter scaffold: motivating situation; definitions and model; developed argument; worked example; results with proofs where warranted; empirical questions; exercises.

Chapter 45

Measurement and Empirical Identification

Translates theoretical quantities into observable indicators and study designs. Maintenance records, switching costs, recovery times, service interruptions, and obligation fulfillment provide evidence. Explains which causal claims require experiments, natural experiments, longitudinal data, or structural assumptions.

Proof programme. P12, P15, P16, P17, P28, P29.

Chapter scaffold: motivating situation; definitions and model; developed argument; worked example; results with proofs where warranted; empirical questions; exercises.

Part X

Categorical Interpretations

Part X interprets economic processes as morphisms and states what the interpretation preserves. Its stated payoff is that parallel composition fails to be independent under shared sinks, and the structure needed to repair that is identified.

HOUSING THREAD. Repair processes become morphisms between building interfaces, and accounting becomes a functor that forgets the backlog.

Chapter 46

Economic Processes as Morphisms

Takes specified economic interfaces or states as objects and admissible transformations, carrying explicit resource and obligation effects, as morphisms. Identity and composition are defined before any categorical claim. The construction begins with the syntactic congruence of the admissible-history language, and where ordinary composition is inadequate the chapter states the extra structure required.

Proof programme. P01, P05, P18.

Chapter scaffold: motivating situation; definitions and model; developed argument; worked example; results with proofs where warranted; empirical questions; exercises.

Chapter 47

Parallel Production and Monoidal Structure

Represents processes performed alongside one another through a monoidal structure. Shared labour, ecological sinks, or scarce machinery complicate the interpretation of independence. String-diagram notation stays optional, and algebraic presentations suffice for the arguments.

Proof programme. P09, P10, P18.

Chapter scaffold: motivating situation; definitions and model; developed argument; worked example; results with proofs where warranted; empirical questions; exercises.

Chapter 48

Resources, Obligations, and Enriched Categories

Motivates enrichment over appropriate ordered structures by costs, delays, risks, and resource requirements. Develops multidimensional cost composition and identifies when a single numerical distance suppresses economically relevant distinctions.

Proof programme. P13, P24.

Chapter scaffold: motivating situation; definitions and model; developed argument; worked example; results with proofs where warranted; empirical questions; exercises.

Chapter 49

Accounting as a Structure-Preserving Map

Connects operational processes to accounting representations by functors. The central question is what information an accounting map preserves, identifies, or loses. Two economically different processes can have the same financial image, which gives a precise form of accounting insufficiency relative to a stated task.

Proof programme. P11, P12, P19, P29.

Chapter scaffold: motivating situation; definitions and model; developed argument; worked example; results with proofs where warranted; empirical questions; exercises.

Chapter 50

Open Systems, Institutions, and Compositional Guarantees

Describes households, firms, and infrastructures through interfaces in an open-systems formalism. Develops an assume-guarantee treatment of institutional composition and identifies conditions under which local guarantees support a global continuation claim. Categorical language organizes the proof obligations while empirical premises remain independently necessary.

Proof programme. P04, P23, P27.

Chapter scaffold: motivating situation; definitions and model; developed argument; worked example; results with proofs where warranted; empirical questions; exercises.

Part XI

Type Theoretic Interpretations

Part XI interprets states, permissions, and commitments as types. Its stated payoff is the precise gap between formal inhabitation and real-world executability, and the separate roles of preservation and progress.

HOUSING THREAD. Repair commitments are typed with deadlines and explicit failure branches, and degraded operating modes carry evidence.

Chapter 51

Economic States and Admissible Actions as Types

Describes states, resources, permissions, and actions with specified preconditions as types. Dependent types express actions whose availability depends on current resources or established authority. Distinguishes formal inhabitation from real-world executability.

$$\mathrm{Act} : \mathrm{State} \to \mathrm{Type}\,.$$

Proof programme. P02, P20.

Chapter scaffold: motivating situation; definitions and model; developed argument; worked example; results with proofs where warranted; empirical questions; exercises.

Chapter 52

Linear and Affine Resources

Models resources that cannot be duplicated freely, or that may be discarded under specified rules, with linear and affine type systems. Money, physical goods, permissions, information, and obligations need different treatments, and no single resource discipline is imposed on all economic objects.

Proof programme. P20.

Chapter scaffold: motivating situation; definitions and model; developed argument; worked example; results with proofs where warranted; empirical questions; exercises.

Chapter 53

Obligations, Deadlines, and Protocols

Expresses exchanges and institutional procedures over time with session types and indexed judgments. Examines escrow, delivery, payment, default, and repair protocols, including explicit failure branches.

Proof programme. P07, P08, P20, P28.

Chapter scaffold: motivating situation; definitions and model; developed argument; worked example; results with proofs where warranted; empirical questions; exercises.

Chapter 54

Proof-Carrying Commitments and Verified Degradation

Commitments carry evidence of specified properties, and degraded operating modes require evidence that required functions remain supported. Small formal examples show how proofs depend on resource models, trusted observations, and assumptions about counterparties.

Proof programme. P03, P06, P14, P16, P20.

Draft note. Candidate import: reconstruction by replay with canonical encoding, parent and tick validation, duplicate-operation handling, and refusal records, stated with the distinction between replay integrity and historical authority.

Chapter scaffold: motivating situation; definitions and model; developed argument; worked example; results with proofs where warranted; empirical questions; exercises.

Chapter 55

Refinement, Interpretation, and the Limits of Formal Assurance

Connects abstract economic obligations with executable checks through refinement types. Distinguishes a verified implementation from a justified specification and explains why formal consistency cannot establish fairness, truthful input data, or ecological adequacy by itself.

Proof programme. P15, P17.

Chapter scaffold: motivating situation; definitions and model; developed argument; worked example; results with proofs where warranted; empirical questions; exercises.

Part XII

Sheaf Theoretic Interpretations

Part XII interprets local knowledge and plans over overlapping contexts. Its stated payoff is the distinction between a presheaf of plans and a sheaf of feasible allocations, together with one genuine cohomological obstruction.

Housing thread. Landlord, city, and contractor each hold local records, and gluing succeeds or fails depending on whether the shared crew is represented.

Chapter 56

Local Economic Knowledge and Overlapping Contexts

Introduces a base of contexts such as jurisdictions, organizations, regions, or information domains. Restriction maps describe how a claim or plan appears within a narrower context. The choice of base is economically motivated and explicit.

Proof programme. P21.

Chapter scaffold: motivating situation; definitions and model; developed argument; worked example; results with proofs where warranted; empirical questions; exercises.

Chapter 57

Presheaves of Plans, Accounts, and Commitments

Local plans and records form presheaves when restriction is well defined. Asks whether accounts agree on overlaps, whether obligations receive consistent treatment across departments, and whether local descriptions omit shared dependencies.

Proof programme. P21.

Chapter scaffold: motivating situation; definitions and model; developed argument; worked example; results with proofs where warranted; empirical questions; exercises.

Chapter 58

Gluing and Global Feasibility

Interprets the sheaf condition as a compatibility-and-reconstruction property. Distinguishes agreement of descriptions from feasibility of the described activity: matching local plans can still compete for a globally scarce resource unless that resource is represented in the model. Examples show when gluing succeeds and when the proposed assignment is only a presheaf.

Proof programme. P09, P21.

Draft note. Repair exercises: a local-repair condition need not produce a global section; a lower-energy imperfect candidate can beat an exact match under a penalized reconstruction; and a circle example must not state contradictory overlap values.

Chapter scaffold: motivating situation; definitions and model; developed argument; worked example; results with proofs where warranted; empirical questions; exercises.

Chapter 59

Obstructions, Inconsistency, and Institutional Boundaries

Investigates failures of global coordination through carefully chosen mathematical models. Cohomological methods are introduced only where an appropriate coefficient structure and a justified obstruction interpretation exist. Distinguishes a genuine mathematical obstruction from a loose analogy about disagreement, beginning with the frictionless exchange graph.

Proof programme. P22.

Chapter scaffold: motivating situation; definitions and model; developed argument; worked example; results with proofs where warranted; empirical questions; exercises.

Chapter 60

Descent, Federation, and the Organization of Collective Futures

Examines how independently governed institutions can coordinate compatible records, rights, and obligations without one institution absorbing the others. Brings the three interpretations together: categorical composition for processes, type theoretic judgments for admissibility, and sheaf theoretic compatibility for local-to-global coordination.

Proof programme. P18, P20, P21, P22.

Chapter scaffold: motivating situation; definitions and model; developed argument; worked example; results with proofs where warranted; empirical questions; exercises.

Appendix A

Prerequisites

The appendices supply background so that Parts I to VIII need no outside reading beyond calculus and elementary probability.

A.1 Optimization

Convex programs, Lagrangian duality, Slater's condition, complementary slackness, and shadow prices (used for P24). Dynamic programming and the Bellman equation for the benchmark cases.

A.2 Dynamical Systems

Discrete-time systems, invariant sets, attractors, and thresholds. Continuous-time viability and Hamilton–Jacobi reachability [3, 14].

A.3 Probability

Transition kernels, almost-sure events, measurability and selection, and the facts about Borel and analytic sets that P25 requires [4].

A.4 Order Theory

Partial orders, lattices, the Knaster–Tarski theorem [17], order dimension, and multi-utility representations [8].

A.5 Category Theory

Categories, functors, monoidal categories, enrichment [11], resource theories [6], and compositional models [9, 12].

A.6 Dependent and Linear Types

Dependent types, linear and affine disciplines [10], session types, and refinement types.

A.7 Sheaves

Presheaves, sites, the sheaf condition, torsors, and Čech cohomology in degree one [13].

Appendix B

The Computational Companion

The directory `companion/` holds a finite-state implementation in Python using only the standard library. Its output is recorded in `companion/EXPECTED_OUTPUT.txt`. The checks below are *reported checks*: they have been run by the author of the skeleton, and the package and recorded output should be inspected before any of them is cited as verified. Run `python3 -m vcc.run all` from `companion/`, or `python3 -m unittest discover tests`.

Three kinds of check are distinguished. An *exhaustive* check enumerates every case of a small model and so establishes the claim for that model only. A *random* check samples systems from a recorded seed. It can find counterexamples and give implementation evidence, but it cannot prove a universal theorem. A *counterexample* check exhibits a specific system in which a hypothesis has been removed.

Result	Claim checked	Kind	Seed	Hypothesis removed
P03	iteration = Knaster–Tarski union	random, 300 systems	1	–
P03	no termination on an infinite chain	counterexample	–	finiteness
P05	local acceptability without viability	exhaustive, 3 states	–	–
P09	each demand fits, the sum does not	random, 500 draws	2	–
P10	$K_{12} = K_1 \times K_2$	random, 200 pairs	3	–
P10	$K_{12} \subsetneq K_1 \times K_2$	counterexample	–	independence
P23	monotone in G, actions, disturbances	random, 300 systems	4	–
P22	potential vs. negative cycle	exhaustive, 3 nodes	–	friction varied
P28	lost-exit interval in closed form	random, 400 parameter sets	5	–
P29	availability, timing, sufficiency	random, 600 cases	6	–

Random checks confirm closed forms that are proved in the text; they are not part of any proof. The housing-case numbers in Theorems H1 and H2 are reproduced exactly by the last two rows.

Appendix C

Framework Concordance

This appendix records which earlier text supplies each concept, how it has been modified for economic use, and which connections remain proposals. Provenance is preserved: where material is attributed to the author or stored in an imported collection, it is cited to its original source and not to the collection. All entries are provisional until the author confirms them.

Concept	Source text	Economic modification	Status
Domain establishment	The Domain and Its Establishment	Chapter 4, P02: establishment as an economic activity	proposal
Distinction and continuation	Distinction and Continuation	Chapter 26: classifications as allocators of futures	proposal
Verification without correspondence	Verification Without Correspondence	Chapter 25, P17: what a check establishes	proposal
Aspect relegation	Aspect Relegation Theory	Chapters 23, 24, P16: attention and return margin	proposal
Residuals and replaced things	Economy of Replaced Things	Chapter 10: disposition duties	proposal
Admissible degradation	source to be assigned	Chapter 12, P06	proposal
Continuous intervention	source to be assigned	Chapter 28: cumulative dependence	proposal
Semantic frameworks	source to be assigned	Chapter 21: task-relative adequacy	proposal

Candidate cases from documentation-level review

Two further reviews read project README files only, without inspecting implementations or running tests. A civic-maintenance verification design supplies a case for Chapters 20, 25, and 30. A commerce system with stock reservations and a transactional outbox supplies a resource-commitment case for P09, P20, and P27 (Chapters 31 and 44). An agent-automation tool bears on Chapter 24. Thematic overlap with the book's concerns establishes no influence or borrowing, and promotional claims such as zero risk or general Sybil resistance are excluded.

Candidate imports from a source review

A targeted review of material in the author's imported repositories identified four imports and several counterexamples. The review was not exhaustive, no files were copied, and implementation claims were not tested.

1. Replay integrity versus historical authority (Chapters 22, 25, 54).
2. The identity $D_{\mathrm{KL}}(P(\cdot \mid B) \,\|\, P) = -\log P(B)$ for restricted distributions (Chapter 16), and the distinction between expenditure and damage to regenerative capacity (Chapters 7, 23).
3. Allocation of finite maintenance capacity through explicit response functions (Chapters 17, 23).
4. Conditional platform revenue $R(q) = (\alpha q + \beta)M(q)$ (Chapter 33), restricted to the narrow revenue proposition.

Counterexamples and exercises: a ledger verifier that does not recompute stored roots or aggregate repeated postings (Chapter 25); a bounded-debt claim corrected by the cumulative-excess condition (Chapter 8); sheaf repair claims that do not yield global sections (Chapter 58). Claims about consciousness, universal entropy laws, cosmological foundations, or unverified performance are not imported.

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